Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part –July 24th 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please comment this sentence: "Design a «good» parallel algorithm by extracting all the available parallelism is **not enough**." (1)

B. Is it possible to eliminate an anti-dependency by modifying the program and ensuring that the program remains equivalent? Is it the same with output-dependency? Please show an example of what can be done in both cases. (1)

C. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=0; i<100; i++)
        for (j=2; j<100; j++)
            a[i][j] = f(a[i][j]) + a[i][j-2];
```

Exercise n. 2

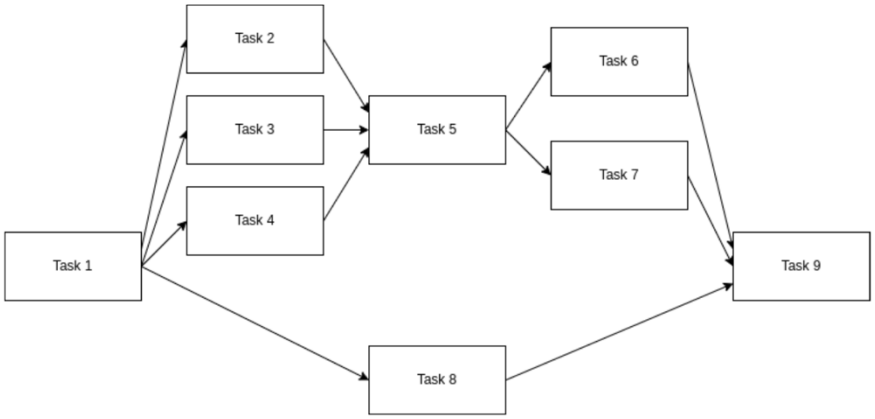
Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the parallel Split pattern: assumptions, complexity, and example. (2)

B. Please describe how the Unzip parallel pattern works. How is it implemented (pseudo code required)? (2)

Exercise n. 3

Answer the questions about the following task graph and briefly explain (without an explanation, the answer will be considered invalid)



W(Task1) = 50,
W(Task2) = 200,
W(Task3) = 250,
W(Task4) = 200,
W(Task5) = 50,
W(Task6) = 100,
W(Task7) = 200,
W(Task8) = 450,
W(Task9) = 50

A. Is it better to implement the program with 3 or 4 threads? (1)

B. Write a corresponding OpenMP implementation. (2)

C. If you could optimize one task bringing its work to 0 (maintaining the graph structure), which one would yield the best improvement in terms of parallel execution time? (1)

Exercise n. 4

Answer the following questions about MPI and briefly explain (without an explanation, the answer will be considered invalid)

A. Why should I use `mpirun` to execute my MPI application? (1) **This question can be skipped if you passed the challenge.**

B. Explain what the `MPI_Comm_split` function does and give an example of its usage, without implementation details such as the function signature. (1)

C. Explain the blocking semantics in point-to-point communications and the problems it introduces in the program. (2)